

1.4.7. Let p be a prime. Prove that order of $GL_2(\mathbb{F}_p)$ is $p^4 - p^3 - p^2 + p$.
 proof. Clearly, the cardinality of $M_{2 \times 2}(\mathbb{F}_p)$ is p^4 .

And, for any $A \in M_{2 \times 2}(\mathbb{F}_p)$,

$\det A = 0$ if and only if $\text{rank } A \leq 1$

if and only if $A = 0$ or $\text{rank } A = 1$

And, the number of $\text{rank } A = 1$ matrices is:

$$\underbrace{(p^2 - 1)}_{\substack{\text{number of non-zero} \\ \text{vectors}}} \cdot \underbrace{p}_{\substack{\text{number of multiples}}} + p^2 - 1 = p^3 + p^2 - p - 1$$

Therefore, $|GL_2(\mathbb{F}_p)| = p^4 - (p^3 + p^2 - p - 1) - 1 = p^4 - p^3 - p^2 + p$.

$$p^4 - \frac{(p^2 - 1) \cdot p + p^2}{\binom{p^2}{p} \cdot p^2}$$

1.4.8. $GL_n(F)$ is non-abelian for $n \geq 2$

Proof: Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

$$AB = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{Now, for each } n \geq 2,$$

$$A_n := \begin{pmatrix} A' & 0 \\ 0 & I \end{pmatrix}, \quad B_n := \begin{pmatrix} B' & 0 \\ 0 & I \end{pmatrix}$$

$$\text{Then } A_n B_n = \begin{pmatrix} AB' & 0 \\ 0 & I \end{pmatrix}, \quad B_n A_n = \begin{pmatrix} BA' & 0 \\ 0 & I \end{pmatrix}.$$

1.4.10. Let $G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{R}, a \neq 0, c \neq 0 \right\}$

1). G is closed under the multiplication.

$$\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \cdot \begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix} = \begin{pmatrix} a_1 a_2 & a_1 b_2 + b_1 c_2 \\ 0 & c_1 c_2 \end{pmatrix}$$

2). G has inverse

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \begin{pmatrix} a^{-1} & 0 \\ 0 & c^{-1} \end{pmatrix}$$

3) $G \leq GL_2(\mathbb{R})$

4). $H = \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} \mid a, b \in \mathbb{R}, a \neq 0 \right\} \leq G$.

1.4.11.

1.7. 17. Let G be a group.

G is abelian if and only if $G \rightarrow G, g \mapsto g^{-1}$ is Homomorphism.

Proof This map is clearly bijection. And, $\forall a, b \in G,$
 $ab = ba \Leftrightarrow a^{-1}b^{-1} = b^{-1}a^{-1}$

Thm proved.

1.7.5 Kernel of Action of a group G on A is the same as the kernel of its representation.
 proof. Let $G \times A \rightarrow A: (g, a) \mapsto g \cdot a$ be an Action. We can consider a Homomorphism:

$$\varphi: G \rightarrow S_A: g \mapsto \sigma_g \text{ where } \sigma_g: A \rightarrow A: a \mapsto g \cdot a$$

By definition, the kernel of given action is a subset of G :

$$\{g \in G \mid g \cdot a = a, \text{ for all } a \in A\}.$$

And,

$$\ker \varphi = \{g \in G \mid \varphi(g) = \text{id}\} = \{g \in G \mid \sigma_g = \text{id}\} = \{g \in G \mid g \cdot a = a, \text{ for all } a \in A\}$$

1.7.6. Recall that: an Action of G on A is called faithful if:
 the induced homomorphism φ is injective.

Prove that group G acts faithfully on A

if and only if

kernel of this action is consisted by only identity.

proof. Trivial.

1.7.8. Let A be a non-empty set (not necessarily finite), and let $K \in \mathbb{N}$ s.t. $K \leq \#(A)$.
 put $\mathcal{B} = \{B \subseteq A \mid \#(B) = K\}$. Define an Action of S_A on $\mathcal{B}$ as:

$$S_A \times \mathcal{B} \rightarrow \mathcal{B}: (\sigma, B) \mapsto \sigma[B]$$

Show that it satisfies Axiom of Action.

proof. Clearly, $\text{id} \in S_A$ satisfies $\text{id}[B] = B$. And, let $\sigma_1, \sigma_2 \in S_A$ be given. Then,

$$(\sigma_1 \circ \sigma_2)[B] = \sigma_1[\sigma_2[B]] \text{ for any } B \in \mathcal{B}$$

Therefore it is actually group action.

Example: Consider $A = \{1, 2, 3, 4\}$ and $\mathcal{B} = \{B \subseteq A \mid \#(B) = 2\}$.

Then, $\sigma = (1\ 2)$ acts on $\mathcal{B}$ as:

$$\sigma \cdot \{1, 2\} = \{1, 2\}, \quad \sigma \cdot \{3, 4\} = \{3, 4\}, \quad \sigma \cdot \{1, 3\} = \{2, 3\}, \quad \sigma \cdot \{2, 3\} = \{1, 3\}$$

$$\sigma \cdot \{1, 4\} = \{2, 4\}, \quad \sigma \cdot \{2, 4\} = \{1, 4\}.$$

Group Action.

Def. Let G be a group, and A be a non-empty set.

A map $\alpha: G \times A \rightarrow A: (g, a) \mapsto g \cdot a$ is called Group Action if:

1). For all $a \in A$, $1_G \cdot a = a$

2). For all $g_1, g_2 \in G$ and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Thm. Let $\alpha: G \times A \rightarrow A$ be a group action. Then,

$$\varrho: G \rightarrow S_A: g \mapsto \sigma_g \quad \text{where } \sigma_g: A \rightarrow A: a \mapsto g \cdot a$$

is Homomorphism. Conversely, if $\varrho: G \rightarrow S_A$ is Homomorphism, then

$$\alpha: G \times A \rightarrow A: (g, a) \mapsto \varrho(g)(a)$$

is group action

7.1.17. Let G be a group. Consider an Action of G on G as:

$$G \times G \rightarrow G; (g, x) \mapsto gxg^{-1}$$

For fixed $g \in G$, $x \mapsto gxg^{-1}$ is Automorphism on G .

Deduce that $|x| = |gxg^{-1}|$ for all $x \in G$, and $|A| = |gAg^{-1}|$, for any $A \subseteq G$.

Proof. Let $y \in G$. Then, $g(g^{-1}yg)g^{-1} = y$. Thus surjection.

And, suppose $gx_1g^{-1} = gx_2g^{-1} \Rightarrow x_1 = x_2$, by cancellation, therefore injective.

Moreover, Homomorphism is given by $gx_1g^{-1} \cdot gx_2g^{-1} = g(x_1x_2)g^{-1}$.

To show $|x| = |gxg^{-1}|$, let $|x| = n \in \mathbb{N}$. Since

$$(gxg^{-1})^n = gx^n g^{-1} = g1g^{-1} = 1, \quad |gxg^{-1}| \leq n.$$

And, if $(gxg^{-1})^k = 1$ for some $k < n$, then

$$(gxg^{-1})^k = gx^k g^{-1} = 1 \text{ implies } x^k = g^{-1}g = 1.$$

which is Contradiction.

To show $|A| = |gAg^{-1}|$, define a map

$$A \rightarrow gAg^{-1}; a \mapsto gag^{-1}.$$

Then, it is bijective (not necessary Homomorphism).

7.1.6. Let G be an abelian group. Show that the Torsion subgroup

$$T \stackrel{\text{def}}{=} \{x \in G \mid |x| < \infty\} \subseteq G$$

is subgroup of G .

Proof. Since the identity has order of 1, $T \neq \emptyset$.

Let $x, y \in T$ be given. Since G is abelian, for $n = |x|$ and $m = |y|$,

$$(xy^{-1})^{nm} = x^{nm} \cdot (y^{-1})^{nm} = (x^n)^m \cdot (y^{-m})^n = 1 \cdot 1 = 1$$

Therefore $xy^{-1} \in T$. By the subgroup Criterion, $T \leq G$.

7.1.11. Let A, B be groups. Then,

$$\{(a, 1) \mid a \in A\}, \{(1, b) \mid b \in B\} \leq A \times B$$

and

$$\{(a, a) \mid a \in A\} \leq A \times A.$$

Proof. Clear.

2.1.4

Give an example s.t. $H \subseteq G$ is a subset of a group with closed under the operation, but it is not a subgroup.

sol). $G = (\mathbb{Q} \setminus \{0\}, \cdot)$ and $H = \{\frac{1}{n} \mid n \in \mathbb{N}\}$

2.1.6. Let G be a group. The subset

$$T = \{g \in G \mid |g| < \infty\}$$

is a subgroup when G is abelian. Gives an example T is not subgroup of G .

sol). $|g_1 g_2| = \text{lcm}(|g_1|, |g_2|) < \infty \quad \forall g_1, g_2 \in T$ if G is abelian.

Let $G = GL_2(\mathbb{Q})$. Then, $A = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \in GL_2(\mathbb{Q})$ have order 2,

but $AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has infinite order. $((AB)^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \dots (AB)^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}, \dots)$

2.2.2. $C_G(Z(G)) = G$. sol) $C_G(Z(G)) = \{g \in G \mid \forall a \in Z(G), gag^{-1} = a\} = G$.

2.2.3. If $A, B \leq G$ with $A \leq B$, then $C_G(B) \leq C_G(A)$.

sol). Let $x \in C_G(B)$. Then, $\forall b \in B, xbx^{-1} = b$, therefore $\forall a \in A \leq B, xax^{-1} = a$. Thus $x \in C_G(A)$.

2.2.6. Let $H \leq G$.

a). Prove $H \leq N_G(H)$ and give an example that it is not true if H is not subgroup.

b). Prove $H \leq C_G(H)$ if and only if H is abelian.

Proof. Let $h \in H$. Then, $hHh^{-1} \subseteq H$ because it is closed under the operation, and $H \subseteq hHh^{-1}$ because $\forall k \in H, h^{-1}kh \in H$ and $h^{-1}kh \in hHh^{-1}$.

Counterexample: take $G = S_3$ and $H = \{(1\ 2), (1\ 3)\}$. Then

$$(1\ 2)H(2\ 1) = \{(1\ 2), (2\ 3)\} \neq H$$

And, $H \leq C_G(H) \Leftrightarrow \forall h \in H, h \in C_G(H)$
 $\Leftrightarrow \forall h \in H, gh = hg \ \forall g \in G$
 $\Leftrightarrow H$ is abelian.

2.2.8. Let $G = S_n$. Find $|G_i|$ where $i \in \{1, \dots, n\}$

sol). Let S_n acts on $\{1, \dots, n\}$. Then, the stabilizer $G_i = \{\sigma \in S_n \mid \sigma(i) = i\}$

2.2.10. Let $H \leq G$ with $|H| = 2$. Prove $N_G(H) = C_G(H)$ and deduce that $N_G(H) = G \Rightarrow H \leq Z(G)$.

Proof. $C_G(H) \leq N_G(H)$ is trivial, because $x \in C_G(H) \Leftrightarrow \forall h \in H, xhx^{-1} = h \Rightarrow xHx^{-1} = H$.

Conversely, since $|H| = 2$, denote $H = \{1, a\}$. Then,

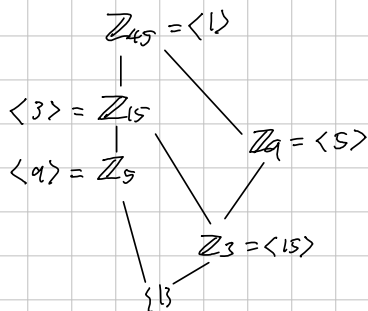
$$x \in N_G(H) \Leftrightarrow xHx^{-1} = H \Leftrightarrow \{1, xax^{-1}\} = \{1, a\} \Leftrightarrow xax^{-1} = a \Rightarrow x \in C_G(H).$$

Thus, if $N_G(H) = G$, then $C_H(H) = G$, thus $H \leq Z(G)$.

2.2.11. $Z(G) \leq N_G(A)$ for any subset $A \subseteq G$.

Proof. Ask a toddler on the street.

2.3.1. Find all subgroup of $\mathbb{Z}_{45}$.



2.3.2. Let G be a finite group. If $|x| = |G|$, then $G = \langle x \rangle$.

But, it fails when G is infinite.

Proof Let $|G| = n$. Then, $1, x, x^2, \dots, x^{n-1}$ are distinct elements in G , therefore $G = \langle x \rangle$. Counterexample: $\mathbb{Q}, \langle 1 \rangle = \mathbb{Z} \subsetneq \mathbb{Q}$.

2.3.12. Prove $\mathbb{Z} \times \mathbb{Z}$ is Not cyclic.

Proof. Suppose $\mathbb{Z} \times \mathbb{Z}$ is cyclic. Let $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ be a generator.

$n \cdot (a, b) = (1, 0)$ for some $n \in \mathbb{N} \Leftrightarrow b = 0$. Thus $(a, b) = (a, 0)$

But, it can not generate $(0, 1)$.

2.3.16. Let $|x| = n$ and $|y| = m$, and x and y commute.

Then $|xy|$ divides $\text{lcm}(n, m)$.

Proof. Let $k = \text{lcm}(n, m)$. Then $(xy)^k = x^k y^k = 1$. Thus $|xy|$ divides k .

Section 4.1.

#1. Let a group G acts on a set A .

$$G_a = \{g \in G \mid g \cdot a = a\}$$

If $a, b \in A$ and $b = g \cdot a$ for some $g \in G$, then $G_b = g G_a g^{-1}$

Deduce that if G acts transitively on A , then kernel of action is $\bigcap_{g \in G} g G_a g^{-1}$

Proof. Let $x \in G_b$ be given. There is, $x \cdot b = b = g \cdot a$ and $x \cdot b = x \cdot (g \cdot a) = xg \cdot a$.

$g \cdot a = xg \cdot a$ implies $a = g^{-1}xg \cdot a$, therefore $g^{-1}xg \in G_a$.

Conversely, let $ghg^{-1} \in g G_a g^{-1}$ be given. Then,

$$ghg^{-1} \cdot b = gh \cdot a = g \cdot a = a.$$

Now, if G acts transitively on A , then

$$\text{Kernel of the action} = \bigcap_{b \in A} G_b = \bigcap_{g \in G} g G_a g^{-1}$$

Section 4.1.

#2. Let A be a set, and let $G \leq S_A$ acts on A . Prove that

For all $\sigma \in G$ and $a \in A$, $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$.

Deduce that if G acts transitively on A , then $\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1$.

Proof. Let $\sigma \in G$ and $a \in A$ be given. Then, by the same problem, $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$.

Meanwhile, the kernel of this action is trivial because

$$\sigma \in \bigcap_{a \in A} G_a \iff \sigma(a) = a \text{ for all } a \in A$$

Now, by the previous problem, $\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1$.

#3. Let $G \leq S_A$ be a abelian, and transitive. Show that

for all $\sigma \in G \setminus \{1\}$ and $a \in A$, $\sigma(a) \neq a$. Deduce that $|G| = |A|$.